Experiment 6

ARP & PPP Configuration

Lab 6 Objectives:

● Observe ARP message exchange and configure PPP with authentication for direct network connections.

Prerequisites: The prerequisites of this lab are-

● Basic networking knowledge (MAC & IP addressing, ARP, PPP).
● Familiarity with command-line tools for network configuration.

Outcomes: By the end of this lab, students will be able to:

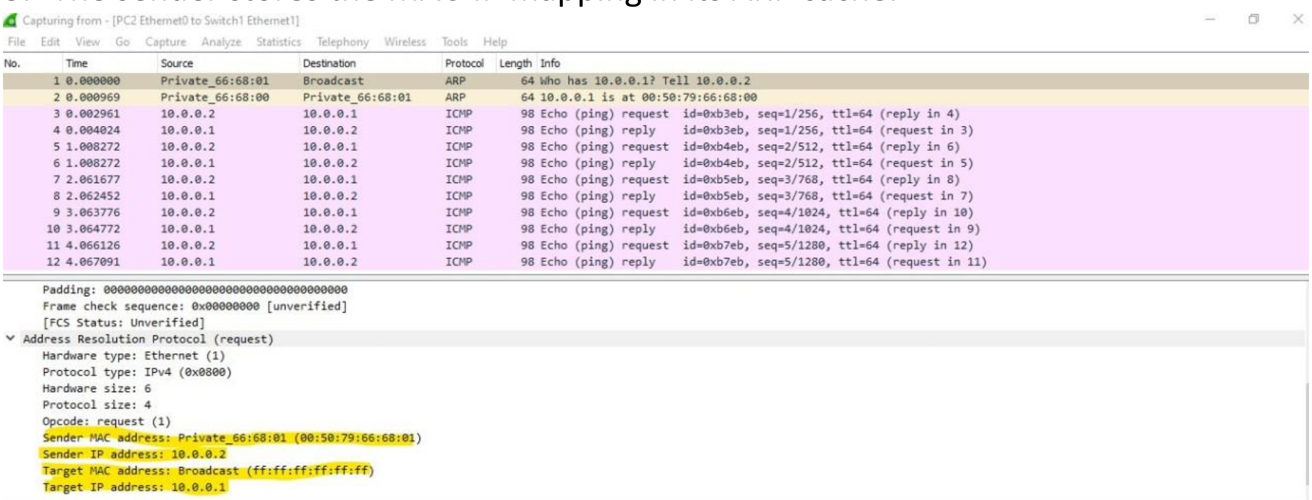
● Capture and analyze ARP messages.
● Configure and verify PPP with authentication.
● Troubleshoot ARP and PPP-related issues.

What is ARP (Address Resolution Protocol)?

ARP is used to map IP addresses to MAC (hardware) addresses in a local area network. It is essential for devices to communicate within the same subnet.

How ARP Works:

1. If a device wants to send data to another device but only has its IP address, it sends an **ARP Request** (broadcast).
2. The target device replies with an **ARP Reply** (unicast), providing its MAC address.
3. The sender stores the MAC-IP mapping in its ARP cache.



What is PPP (Point-to-Point Protocol)?

PPP is a data link layer protocol used to establish a direct connection between two networking nodes (usually routers) over serial links.

Features of PPP:

- Supports authentication using PAP or CHAP.
- Encapsulates network layer protocols.
- Error detection and compression.

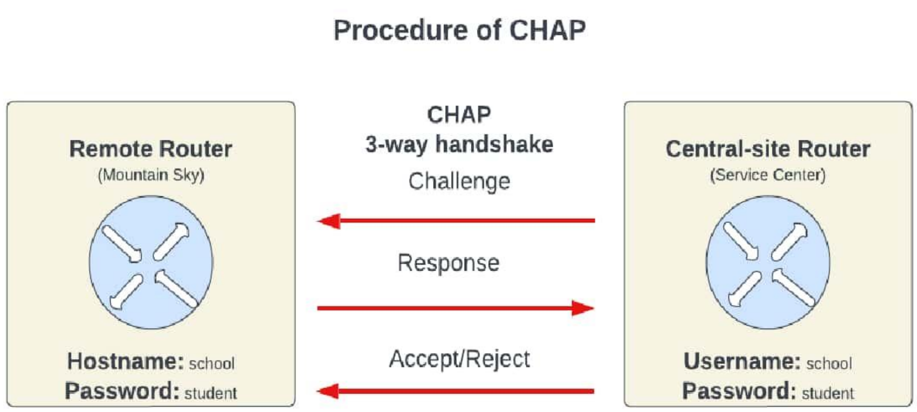
PPP Authentication Methods:

1. PAP (Password Authentication Protocol):

- Two-way handshake.
- Transmits username and password in plain text.
- Less secure.
-

2. CHAP (Challenge Handshake Authentication Protocol):

- Three-way handshake.
- Encrypts the password using a hash function.
- More secure.

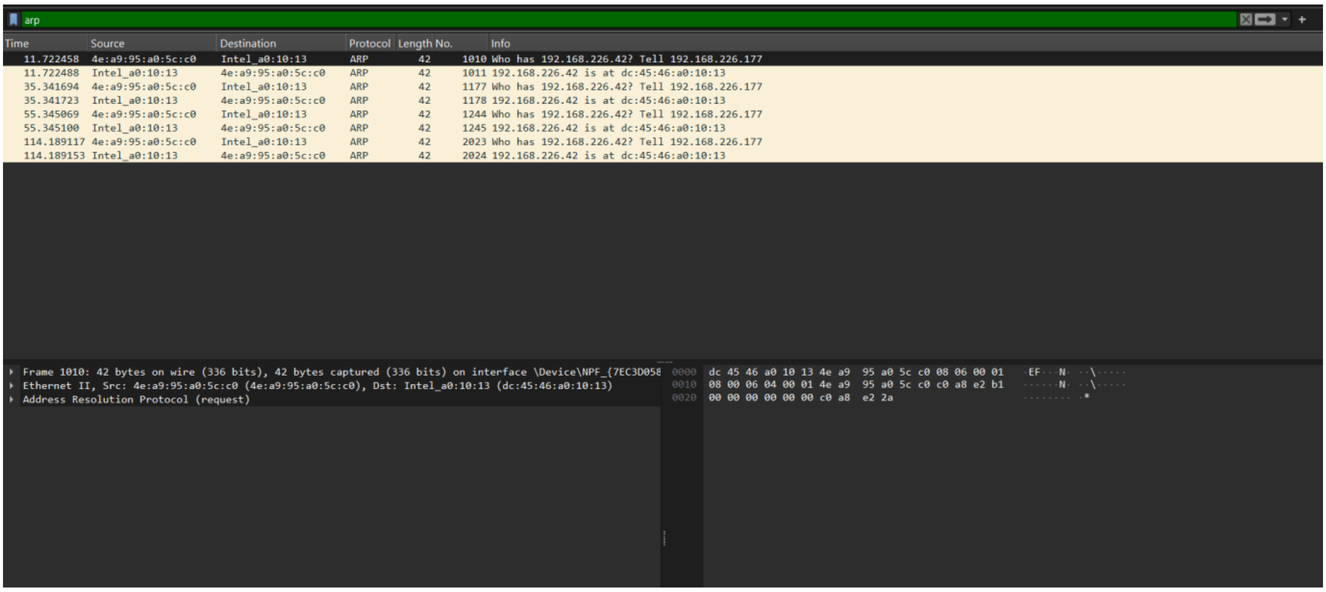


On **PC1 (my laptop)**, I used the ping command to initiate communication with another device in the same subnet:

```
C:\Users\CHETAN>ping 192.168.226.42

Pinging 192.168.226.42 with 32 bytes of data:
Reply from 192.168.226.42: bytes=32 time<1ms TTL=128
Reply from 192.168.226.42: bytes=32 time<1ms TTL=128
Reply from 192.168.226.42: bytes=32 time<1ms TTL=128
Reply from 192.168.226.42: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.226.42:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\CHETAN>
```

This confirms that the destination device is reachable, and the system successfully resolved the IP address using ARP.

Then, using **Wireshark**, I captured the ARP message exchange. I applied the filter:



Observation:

ARP Request: *Who has 192.168.226.42? Tell 192.168.226.177*

ARP Reply: *192.168.226.42 is at dc:**65:56:a0:19:93***

Configure PPP Between Two Routers – Theory

Objective:

To configure a **Point-to-Point Protocol (PPP)** connection between two routers using serial interfaces for direct communication.

What is PPP?

Point-to-Point Protocol (PPP) is a **Data Link Layer (Layer 2)** protocol that is used to establish a **direct connection between two networking devices** such as routers over serial links. PPP supports:

- **Encapsulation of network layer protocols** (like IP, IPX)
- **Authentication mechanisms** (PAP and CHAP)
- **Compression and error detection**

Why Use PPP?

Although routers by default use **HDLC (High-Level Data Link Control)** as their encapsulation protocol on serial interfaces, PPP is more versatile and supports

- **Multivendor compatibility**
- **Advanced authentication (PAP/CHAP)**
- **Link quality monitoring**
-

Therefore, PPP is widely used in WAN (Wide Area Network) links where authentication and reliability are important.

Required Setup:

To configure PPP, we need:

- Two routers with **serial interfaces**
- A **serial (DCE/DTE) cable** to connect the routers
- IP addresses on both interfaces that belong to the **same subnet**

Configuration Commands:

On Router1:

```
interface Serial0/0/0
ip address 10.0.0.1 255.255.255.252
encapsulation ppp
no shutdown
```

On Router2:

```
interface Serial0/0/0
ip address 10.0.0.2 255.255.255.252
encapsulation ppp
no shutdown
```

Explanation:

- interface Serial0/0/0: Enters the serial interface configuration mode.
- ip address: Assigns an IP address in the same subnet to the interface.
- encapsulation ppp: Changes the default encapsulation from HDLC to PPP.
- no shutdown: Activates the interface.

Expected Outcome:

After configuring PPP on both routers:

- The serial link becomes **active**.
- PPP is used for encapsulation.
- The routers can **communicate** with each other using IP over a **PPP link**.
- This setup can later be extended to add **authentication (CHAP/PAP)**.

Analysis Report

Experiment Title:

Observing ARP & Configuring PPP Authentication

The primary objective of this experiment was to:

- Understand the working of Address Resolution Protocol (ARP) in mapping IP addresses to MAC addresses within a local network.
- Configure and test Point-to-Point Protocol (PPP) with authentication methods like CHAP to secure communication over serial links.

Findings:

- ARP Functionality:
  The experiment demonstrated that ARP plays a vital role in allowing devices to communicate over a LAN by translating IP addresses into MAC addresses. This translation is essential for packet delivery within the same network.
- PPP with CHAP Authentication:
  The PPP configuration using CHAP (Challenge Handshake Authentication Protocol) ensured a more secure authentication process between two routers. The CHAP method involves a three-way handshake and periodically verifies the identity of the peer.

Problems Faced:

1. Authentication Failure:
   Initially, PPP authentication using CHAP failed because the username and password were mismatched between the two routers.

2. Missing ARP Packets:
   ARP requests and replies were not visible during packet capture when the network cable was disconnected, causing devices to fail in resolving MAC addresses.

Solutions Implemented:

1. Correcting Authentication Configuration:
   The username and password were rechecked and updated to ensure both routers had matching credentials for CHAP authentication.

2. Enabling Interface:
   The serial interface was reactivated using the **no shutdown** command, restoring connectivity and allowing ARP packets to be exchanged successfully.